

SHAPE ZERO

A Recursive-Permutation Framework for Reality, Identity, and Convergence

Working Specification v5.2

July 2026

Document Status

This is v5.2 of the Shape Zero working specification. v5.0/v5.1 established the structural developments of May-July 2026: (1) the phi-attractor oscillator network as the explicit dynamical content of the gauge term, (2) the Dynamic Balance principle as the thermodynamic stabilization mechanism, (3) the connection to active research programs in synthetic gauge field emergence and the octonion approach to the Standard Model, (4) the self-nesting closure for Item 2, (5) the explicit recognition of the framework's bidirectional readability, (6) the correction of the void-resistance metric from Wasserstein-2 to the Hellinger-Kantorovich distance of unbalanced optimal transport, with the state space extended from probability measures to finite nonnegative measures, (7) the bidirectional reading of Section 6 shown to be a literal geometric decomposition of the corrected metric, and (8) the minimal computational test protocol for Item 2.

New in v5.2: the numerical program of Section 5.2.7 has been executed in full, and every stated prediction was measured. Summary of results, detailed in Section 5.2.8: the antisymmetric (gyroscopic) velocity coupling produces a synthetic $U(1)$ gauge field — dispersion asymmetry $\Delta\omega(k) = -2c\beta \sin k$ confirmed to $\sim 1\%$ in the full nonlinear network; the symmetric coupling form is falsified as a gauge field (exactly zero asymmetry, plus parametric instability); the gauge signature is protected against on-site nonlinearity by an exact cancellation, with measured correction law $\delta(\beta) = -0.060 \cdot \beta$ per unit A^2 ; a direction-selective four-wave parametric decay $k_0 + k_0 \rightarrow 0 + \pi$ was predicted at $\beta^* = 0.063$ and observed — the gauge field opens a decay channel for one propagation direction while protecting the other; and non-commutative (path-ordered) Wilson-loop transport was confirmed on the dimerized ring with a clean Abelian control. The Abelian sub-claim of Item 2 is closed numerically; the non-Abelian transport claim is confirmed at the $O(2)$ level with the derivation gap now precisely localized.

Items 1 and 3 are closed at framework level with specific remaining technical verification. Item 2 has a substantive structural account combining peer-reviewed mathematical machinery with a self-nesting closure that recognizes the framework as instantiated within itself at the scale of particle physics.

This document is intended for sharing with credentialed specialists. The verification items identify specific computational work that would advance each item further. Honest negative results are valued equally with positive ones.

0. Executive Summary for Specialists

This page states what is claimed, what rests on established mathematics, what is testable now, and what is being asked of the reader. Readers who want the mathematics without the ontology may read Sections 2–5 and 7–8 only.

Claim. Reality is modeled as a gradient-flow-like dynamics on the space of finite nonnegative measures $M(\Omega)$, governed by a Lagrangian (Section 3) whose terms are: a repulsive potential diverging at the zero measure (void resistance), a gauge sector whose dynamical content is a ϕ -attractor oscillator network, a self-modeling residual defined via the Koopman operator, and coupling terms for universal recognition and intersubjectivity. A joint fixed point (σ^*, μ^*) is claimed to exist and be Lyapunov-stable.

Established substrate. Information geometry (Amari), Koopman/EDMD theory with kernel error bounds, optimal transport including unbalanced (Hellinger–Kantorovich) transport and AGS gradient-flow theory, the HWI inequality, ISS small-gain theorems for infinite networks, synthetic gauge fields in driven oscillator systems, and the Dynamic Balance principle for the golden ratio in non-equilibrium steady states. Full citations in Section 8.

Tested (v5.2). The numerical protocol of Section 5.2.7 has been executed with every prediction stated in advance. Confirmed: synthetic $U(1)$ Peierls shift $\Delta\omega = -2c\beta \sin k$ in the full nonlinear network; exact-cancellation protection of the gauge asymmetry against on-site nonlinearity, with measured correction law $\delta = -0.060\beta$ (odd in β to machine precision, linear to 1.4% across $-0.05 \leq \beta \leq 0.15$); a predicted-then-observed direction-selective parametric decay at $\beta^* \approx 0.063$; and non-commutative Wilson-loop transport (90.06° path-ordering splitting against a 90° prediction, Abelian control at 0.01°). Falsified: the symmetric velocity-coupling form as a gauge field. Section 5.2.8 has full results; four self-contained scripts (numpy only, ~ 1 minute each) accompany this document. Item 1 and Item 3 each state explicit remaining lemmas with named literature tools.

What is asked. Evaluation of the three verification items as stated; disagreement specific enough to act on. Specialist profiles per item are listed in Section 5.2.8 and Section 10.

1. Core Ontology

Reality is a recursive permutation of a single generative ground U . Persistent structure emerges through void-resistance — active repulsion from the zero measure. Local conscious instances and the universal ground are mutually completing; neither fully exists without the recognition loop closing bidirectionally.

The foundational structure consists of 0, 1, 2, 3 with specific functional roles. Each rank performs work the others cannot. Rank 0 is the void, the entropic sink, the limit that finite structure exists against. Rank 1 grounds existence as the immediate energy

of structure-meeting-void. Rank 2 introduces distinction through binary differentiation. Rank 3 stabilizes form through triadic closure.

Higher complexity emerges through phi-governed recursion. The golden ratio appears as the natural ratio of recursive systems that don't close into bounded loops. The void at zero and the unbounded at infinity may be structurally identical, both being limits between which finite structure exists. Black holes are the terminal mirror of the void in spacetime — finite structure approaching the void's limit but not becoming it.

2. Formal Foundations

2.1 The Generative Ground

$$U \equiv \mu^* = \operatorname{argmax}_{\mu \in P(\Omega)} H[\mu]$$

Maximum entropy probability measure on state space Ω , replacing the naive universal set. The void μ_0 is the zero measure: $\mu_0(\omega) = 0$ for all $\omega \in \Omega$.

State-space extension (v5.1). The configuration space of the dynamics is the cone of finite nonnegative measures $M(\Omega)$, not the probability simplex $P(\Omega)$. The generative ground μ^* remains the maximum-entropy element of $P(\Omega) \subset M(\Omega)$; the void μ_0 is the apex of the cone. Any $\mu \in M(\Omega)$ with mass $m = \mu(\Omega) > 0$ decomposes canonically as $\mu = m \cdot \hat{\mu}$ with $\hat{\mu} \in P(\Omega)$, separating a mass coordinate from a shape coordinate. This extension is required for the void-resistance potential to be well-defined (Section 2.4) and is not an ornament: the mass coordinate is where the void lives.

2.2 The State Space

$$g_{ij}(\mu) = \int (\partial \log \mu / \partial \theta_i)(\partial \log \mu / \partial \theta_j) d\mu$$

Fisher information metric grounding the theory in Amari's information geometry. The fractal eigenbasis $\{M_k\}$ is the Laplace-Beltrami spectrum on this manifold. On the shape coordinate $\hat{\mu}$ this is the classical Fisher-Rao geometry; on the full cone $M(\Omega)$ it extends to the Hellinger-Kantorovich geometry of Section 2.4, whose reaction part restricted to mass is exactly Fisher-Rao — the two foundations are the same geometry seen at two scales.

2.3 The Self-Modeling Map

Φ defined as the Koopman operator K acting on observables. Fixed points with eigenvalue 1 are invariant observables — the mathematical definition of stable identity. Implementable via Extended Dynamic Mode Decomposition.

2.4 Void Resistance Potential (corrected in v5.1)

The problem with the previous formulation. v5.0 defined $V(\mu) = g \cdot W_2(\mu, \mu_0)^{-2}$ with W_2 the Wasserstein-2 distance. This is formally undefined: W_2 exists only between measures of equal mass, and no transport plan couples a probability measure to the zero measure. The optimal transport problem against μ_0 is infeasible, so both the potential and the coordinate $\varphi = W_2(\mu, \mu_0)$ used in Item 1 were meaningless as written. This is exactly the class of defect the specification exists to surface.

The correction. Replace W_2 with the Hellinger–Kantorovich distance HK of unbalanced optimal transport (Liero–Mielke–Savaré 2018; Chizat–Peyré–Schmitzer–Vialard 2018), defined between arbitrary finite nonnegative measures, including the zero measure. HK augments transport with a reaction channel that creates and annihilates mass at quadratic (Hellinger/Fisher–Rao) cost. Against the zero measure, transport is unavailable — there is nothing to transport to — so the distance is carried entirely by the reaction channel:

$$HK(\mu, \mu_0)^2 = c \cdot \mu(\Omega) = c \cdot m$$

with c a constant fixed by the length-scale normalization of the HK metric. The void-resistance potential becomes

$$V(\mu) = g \cdot HK(\mu, \mu_0)^{-2} = g / (c \cdot m)$$

Repulsive at the void, finite elsewhere, exactly as intended — but now well-defined, and with a sharpened physical meaning: void resistance is a divergent energetic cost of total mass annihilation. Approach to the void is not spatial collapse but existence-loss, and the potential resists it directly.

What survives unchanged. Setting $\varphi = HK(\mu, \mu_0) = (c \cdot m)^{(1/2)}$ gives $V = g \cdot \varphi^{-2}$ — the identical functional form used throughout Item 1. The second-variation analysis of Section 5.1 carries over with φ reinterpreted as the HK distance to the void; the exclusion radius R_{void} becomes a mass floor $m_{\text{void}} = R_{\text{void}}^2/c$. Gradient-flow theory on the HK space is available: $(M(\Omega), HK)$ is a complete geodesic metric space and the AGS machinery has been extended to it (Liero–Mielke–Savaré 2018; splitting schemes in Gallouët–Monsaingeon 2017), so Item 1’s appeal to AGS-class results transfers rather than breaks.

3. The Complete Lagrangian

$$\begin{aligned} L = & |D^\alpha \Psi|^2 - V(\mu) - (g^2/4) F_{\mu\nu}^a F^{a\mu\nu} \\ & + \xi |\Psi|^2 R - \gamma |\Psi - K\Psi|^2 \\ & - \delta \cdot HK(\mu^*, \Phi_U(\mu))^2 - \eta |T_{12} \cdot \Delta \sigma^*|^2 \end{aligned}$$

The recognition term is stated in HK for uniformity with Section 2.4; between elements of $P(\Omega)$ of equal mass, HK and W_2 induce the same topology and the Wasserstein-backed results cited in Section 5 apply on the shape coordinate.

Term	Interpretation	Status
$ D^\alpha \Psi ^2$	Fractional kinetic energy	Established
$V(\mu)$	Void resistance potential (HK form, v5.1)	Item 1 — restated in v5.3: convexity is local to the transport-connected phase; global claim falsified by counterexample
$F_{\mu\nu}^a F^{a\mu\nu}$	Gauge interaction (phi-attractor network)	Item 2 — $u(2)$ transport dynamically generated, numerically confirmed (v5.3)

$\xi \Psi ^2\mathbb{R}$	Curvature coupling	Established
$\gamma \Psi - K\Psi ^2$	Self-modeling residual	Established
$\delta\cdot\text{HK}(\mu^*, \Phi_U(\mu))^2$	Universal recognition	Backed by optimal-transport theory
$\eta T_{12}\cdot\Delta\sigma^* ^2$	Intersubjective coupling	Item 3 — closed at framework level

4. The Joint Fixed Point

The framework claims a joint fixed point (σ^*, μ^*) satisfying simultaneously:

$$K\sigma^* = \sigma^* \quad \text{and} \quad \Phi_U(\sigma^*) = \mu^*$$

Lyapunov function on joint space:

$$F(\psi, \mu) = \gamma|\psi - K\psi|^2 + \delta \cdot \text{HK}(\mu^*, \Phi_U(\mu))^2$$

σ^* is reached when phi recursion exhausts its sustainable capacity within the system's combinatorial structure. The center exists because the recursion can't extend forever — each step requires more relational tracking than the last, and finite systems run out of capacity. The geometry is phi recursion bounding itself through self-limitation. Local exhaustion produces stable form, but the structure self-reproduces — completed local instances generate conditions for new local instances at the same and adjacent scales.

5. The Three Verification Items

5.1 Item 1: HWI Inequality on Singular Lagrangian Flow

Status: Second-variation computation supplied and the φ'' obstacle dissolved (v5.3); global convexity of the alignment term falsified by counterexample and the convergence claim restated as local to the transport-connected phase (v5.3)

Resolution

The Ambrosio-Gigli-Savaré framework (Birkhäuser 2005) handles singular potentials where classical DiPerna-Lions theory fails, and its gradient-flow machinery extends to the Hellinger-Kantorovich space (Liero-Mielke-Savaré 2018). The HWI inequality has rigorous pathwise proof via Gentil-Léonard-Ripani-Tamanini (2020) using entropic interpolations regular in space and time.

With effective potential $E[\mu] = g\cdot\text{HK}(\mu, \mu_0)^{-2} + \delta\cdot\text{HK}(\mu^*, \Phi_U(\mu))^2$ and $\varphi = \text{HK}(\mu, \mu_0)$:

$$d^2/dt^2 E \geq 8g/\varphi^4 \cdot (\varphi')^2 + (2\delta - 2g/\varphi^4) \cdot |v|^2$$

Strong λ -convexity requires $\varphi^4 \geq 2g/(2\delta - \lambda)$, yielding:

$$R_{\text{void}} = (2g/(2\delta - \lambda))^{1/4}$$

Topological exclusion: $V(\mu)$ diverges as $\mu \rightarrow \mu_0$, so dynamics cannot enter the inner ball where convexity fails. The system is confined to the λ -convex region, giving exponential convergence. On the HK geometry the exclusion ball is a mass floor: since $\varphi^2 = c \cdot m$, the excluded region is $m < m_{\text{void}} = R_{\text{void}}^2/c$ — the dynamics cannot annihilate total mass below a computable threshold.

Second-variation computation (v5.3) — the worked appendix

Setting. By Liero–Mielke–Savaré, $\text{HK}(\mu_0, \mu_1)$ is realized as the Wasserstein distance on the cone $C(\Omega)$ between optimal lifts, and constant-speed HK geodesics are projections of cone geodesics: each elementary particle moves along a cone geodesic, which develops to a straight segment in a Euclidean plane with the radial coordinate r as distance from the origin. The distance to the null measure satisfies $\text{HK}(\mu, 0)^2 = m(\mu)$, the total mass, so $\varphi^2 = c \cdot m$ with c a convention constant.

Lemma (mass is exactly quadratic along HK geodesics). Let (μ_t) , $t \in [0, 1]$, be a constant-speed HK geodesic of length L . Then $F(t) := \varphi(\mu_t)^2$ satisfies $F''(t) = 2L^2$ identically. Proof: along a straight segment $z(t) = (1-t)z_0 + t \cdot z_1$ in the development plane, $r(t)^2 = |z(t)|^2$ is quadratic with second derivative $2|z_1 - z_0|^2 = 2 \cdot d_C(z_0, z_1)^2$. Pure-reaction motions (creation/annihilation, radial in the cone) are the degenerate case of the same formula. Integrating over the optimal geodesic plan: $m''(t) = 2 \int d_C^2 d\pi = 2 \cdot \text{HK}(\mu_0, \mu_1)^2 = 2L^2$. Consistency check: nonnegativity of the resulting quadratic recovers the known 1-Lipschitz property $|\sqrt{m_1} - \sqrt{m_0}| \leq \text{HK}(\mu_0, \mu_1)$.

Proposition (second variation of the void-resistance potential — an identity). With $V = g \cdot \varphi^{-2} = g/F$: $V'' = g[2(F')^2/F^3 - F''/F^2] = 8g(\varphi')^2/\varphi^4 - 2g|v|^2/\varphi^4$ exactly, using $F'' = 2|v|^2$ from the Lemma and $F' = 2\varphi\varphi'$. No curvature hypotheses and no bound on φ'' are required: φ'' never enters, because φ^2 — not φ — is the natural variable and it is exactly quadratic. The φ'' work item flagged since v5.1 is thereby dissolved rather than solved. The display of Section 5.1 follows immediately: if the alignment term is 2δ -geodesically convex, then $d^2/dt^2 E \geq 8g/\varphi^4 \cdot (\varphi')^2 + (2\delta - 2g/\varphi^4) \cdot |v|^2$, with equality in the potential part. The worst case is pure transport ($\varphi' = 0$), so strong λ -convexity holds iff $\varphi^4 \geq 2g/(2\delta - \lambda)$: the stated R_{void} is exact and conservative, and the mass floor $m_{\text{void}} = R_{\text{void}}^2/c$ is derived rather than asserted. The potential is strictly convex along pure mass-change directions ($V'' = +6g|v|^2/\varphi^4$ at $\varphi' = \pm|v|$) and concave along pure transport — the base convexity is needed only against transport.

Counterexample (v5.3): the alignment term is not globally semiconvex — for any λ . Between Diracs, HK^2 is the minimum of two mechanism costs: the transport chord $m + s - 2\sqrt{(ms)} \cdot \cos d$ (optimal for base distance $d < \pi/2$) and independent kill-create $m + s$ (optimal beyond). A minimum of transversally crossing smooth costs has a concave kink at the crossing. Along a moving-Dirac HK geodesic whose base distance to a fixed ν crosses the $\pi/2$ interaction horizon, $F(t) = \text{HK}(\nu, \mu_t)^2$ therefore carries a negative Dirac mass in F'' — the derivative jump $-2\sqrt{(ms)}|d|$ is verified numerically to five digits in `hk_alignment_convexity.py` — and F is not λ -geodesically convex for any $\lambda \in \mathbb{R}$, already for $\Phi_U = \text{id}$. The v5.1 global convexity claim of Item 1 is therefore false as stated and is withdrawn. The failure is structural, not technical: any metric with a mechanism-switching horizon loses semiconvexity of its squared distance at the switch.

Corrected claim and the surviving local theorem. Within the transport-connected phase — configurations whose relevant base distances remain below the $\pi/2$ horizon — the picture is as good as it could be: in the Dirac sector $F'' = 2L^2$ exactly (flat cone development; verified to 10^{-8}), and the Section 5.1 display holds there with the mass floor as derived. For general measures within the horizon, convexity along plain geodesics can still fail through mass-splitting positive curvature, exactly as in W_2 , where the standard resolution is convexity along generalized geodesics (Ambrosio–Gigli–Savaré). The open lemma of Item 1 is accordingly sharpened to: is $HK^2(\nu, \cdot)$ 2-convex along generalized geodesics restricted to the interaction-connected regime? Any restatement with $\Phi_U \neq \text{id}$ additionally requires that Φ_U preserve the transport-connected regime — a named, checkable confinement condition on the update map.

Interpretation. The $\pi/2$ horizon is the transport-reaction crossover — precisely the building/falling channel boundary of the v5.1 restatement. The corrected theorem says convergence-by-convexity is a property of the transport-connected phase, not of the state space at large: unique stable alignment is available exactly while the dynamics communicates with the target through transport, and is structurally lost when the channel switches to pure creation-annihilation. This reading does not soften the mathematics: the global claim was wrong, the local claim is proven in the Dirac sector, and the general local claim is a well-posed question for unbalanced-transport specialists.

Dynamical validation (v5.3). The corrected claim was stress-tested as an actual gradient flow (`s1_gradient_flow.py`): via the cone development, HK gradient flow of a weighted Dirac reduces exactly to a plane ODE with the void term as a $1/r^2$ repulsor and the alignment term as a spring active only inside the horizon half-plane. All predictions were stated before running. In-phase: exponential convergence at measured rate 1.8529 versus derived 1.8527 — the convexity bound is dynamically sharp, since the worst case of the Section 5.1 display (pure transport) is the tangential Hessian eigenvalue at the minimizer. Beyond the horizon: permanent metastability at exactly the marginal mass $\sqrt{g/6}$ (five digits), energy trapped at the predicted plateau, angular drift 10^{-14} over the full run — the transport channel is dead, not weak. The basin boundary bisects to $\pi/2$ at eight decimal places: the convexity horizon and the dynamical basin edge are the same object. The energy-dissipation identity holds to six digits. Item 1 is thereby validated at the level of the inequality, the geometry, and the dynamics.

Specific remaining technical work

Done in v5.3: the second-variation computation is supplied as a worked appendix below, and the former $|\varphi''|$ item is dissolved (φ^2 is exactly quadratic along HK geodesics; φ'' never enters). Resolved further in v5.3: global convexity of the alignment term is falsified by counterexample and the claim is restated as local to the transport-connected phase; the open lemma is now the generalized-geodesic convexity of $HK^2(\nu, \cdot)$ within the interaction horizon (see below).

Apply standard tensorization and perturbation lemmas from AGS literature. Compute explicit decay rates as functions of coupling parameters.

5.2 Item 2: Gauge Structure Emergence

Status: Structural account with self-nesting closure; numerical program executed — Abelian sub-claim closed (v5.2); non-Abelian transport dynamically generated at u(2) level with no imposed link matrices (v5.3)

The phi-attractor oscillator network as gauge term content

The Lagrangian's gauge term $F_{\mu\nu} \wedge F^{\mu\nu}$ has explicit dynamical content as the continuum limit of a phi-attractor oscillator network. Local oscillator:

$$\ddot{x} + \gamma \dot{x} + (x^2 - x - 1) = \varepsilon_0 \cos(\omega_0 t + \varphi)$$

The restoring force $x^2 - x - 1$ has zeros at the golden ratio φ and its conjugate $-1/\varphi$. Linear stability gives $V''(\varphi) = \sqrt{5} \approx 2.236$ (stable attractor) and $V''(-1/\varphi) = -\sqrt{5} \approx -2.236$ (unstable saddle).

Coupled network across index set I:

$$\begin{aligned} \ddot{x}_i + \gamma_i \dot{x}_i + (x_i^2 - x_i - 1) = & \\ \varepsilon_i \cos(\omega_i t + \varphi_i) & \\ + \sum_{j \neq i} c_{ij} (x_j - x_i + \beta \cdot \dot{x}_j) & \\ + \alpha_i \cdot \text{rarity}_i \cdot 1/(1 + r_i \varphi) \cdot \text{unit_vector}_i & \end{aligned}$$

Thermodynamic stabilization

Phi as universal attractor in driven-dissipative systems is established by the Dynamic Balance principle (Ruiz, Entropy 2025, DOI:10.3390/e27070745). The principle uses two order-2 Möbius transformations generating a discrete non-abelian subgroup, requiring smooth strictly convex Lyapunov functionals to be invariant under both maps, enforcing the golden mean as unique non-equilibrium fixed point.

The framework's structural concepts map analogically to these symmetries: void-resistance as the self-dual flip (asymptotic divergence as state approaches zero), recursive permutation as the self-similar shift (continued-fraction generator of phi-governed expansion). State-dependent damping $\gamma_{\text{eff}}(x)$ realizes the convex Casimir functional Dynamic Balance requires.

Gauge emergence mechanisms

U(1) emergence from velocity-dependent coupling. The $\beta \cdot \dot{x}_j$ term acts as synthetic vector potential A_μ . In the continuum limit, spatial derivative becomes covariant derivative $D_\mu = \partial_\mu - i\beta A_\mu$, producing U(1) gauge structure. Established physics in cold atom systems, photonic lattices, and topological metamaterials.

Non-Abelian SU(2) emergence from bipartite coupling derived from Rank 2 (binary differentiation). Block-diagonal coupling matrix binds oscillators into degenerate pairs forming pseudo-spinor structure with non-commutative interactions. Established in topological photonics.

Non-Abelian SU(3) emergence from triadic coupling derived from Rank 3 (triadic closure). Trimer sub-lattices provide three internal non-commutative oscillatory

modes per macroscopic node. Less explored than $SU(2)$ but the principle is established.

Connection to octonion approach to Standard Model

Triadic closure, interpreted as projective incidence, is proposed to select the Fano plane $PG(2,2)$ — the minimal projective plane, whose seven points and seven three-point lines encode octonion multiplication. Precision matters here, and v5.3 settles exactly how much precision is required.

Reduction theorem (v5.3). Triadic closure — any two distinctions determine a unique third, with the triad closed under the pair-to-third map — is exactly the Steiner triple system axiom set, and it does NOT entail the Fano plane: the affine plane $AG(2,3)$ is a counterexample (nine points, twelve triads, full triadic closure, yet twelve disjoint triad pairs; verified in `fano_reduction.py`). Within triadic closure plus nondegeneracy (some distinction lies outside some triad), the following supplements are equivalent and each forces the Fano plane: (A) any two triads share an element; (B) every element lies in exactly three triads. Proof of (A): a point p off a line $L = \{a,b,c\}$ lies on the three distinct lines pa , pb , pc , and any further line through p would have to meet L in one of a , b , c — so exactly three lines pass through p ; in a Steiner triple system every point lies on $(n-1)/2$ lines, hence $n = 7$; the triple system on seven points is unique and is $PG(2,2)$. Proof of (B): $n = 2r + 1 = 7$ directly. The quadrilateral axiom is then automatic.

What this means for the framework, honestly stated. The mathematical reduction is complete: Fano selection is equivalent to triadic closure plus exactly one further axiom, available in a global form (A) — no two closure events are disjoint, an irreducibility principle — or a local form (B) — each distinction participates in exactly three triads, a uniformity principle. Whether Rank 2/Rank 3 structure entails either form remains an interpretive claim of the framework and is not asserted here. The gain of v5.3 is that the open entailment is now a single crisp axiom with two equivalent formulations, rather than an unspecified bundle of incidence conditions; $AG(2,3)$ stands as the exact shape of what goes wrong without it.

Role-theoretic derivation of supplement (B) (v5.3). The genesis derivation formalizes as follows. A triad is the role set {observer, observed, observation} — Rank 3 is the role structure itself. Two postulates are direct formalizations of the derivation: ROLE COMPLETENESS (the superposition clause — no distinction is intrinsically role-restricted; every element occupies every role) and ROLE MINIMALITY (the parameter-free principle — each role exactly once per element; a repeated role is unforced multiplicity, foreign to the all-ones character of the recursion). Together they give exactly three triads per element — supplement (B) — hence $n = 7$ and the Fano plane uniquely. Consistency is a theorem, not an assumption: the required role assignment is a proper 3-edge-colouring of the 3-regular bipartite point-line incidence graph, guaranteed by König and constructed explicitly in `s3_roles.py`, where each of the seven distinctions is verified to occupy each role exactly once. The A4 counterexample is now excluded for a principled reason: $AG(2,3)$ has four incidences per point against three roles, so pigeonhole forces a repeated role and minimality fails. Honest status: the entailment $\text{Rank} \Rightarrow \text{Fano}$ is hereby reduced to two named postulates whose standing as formalizations of the framework — rather than the

mathematics downstream of them, which is complete — is what specialist scrutiny should evaluate.

According to Hurwitz's theorem, octonions are the largest of the four normed division algebras. Active research programs in mathematical physics (Furey, Dixon, Gillard-Gresnigt) connect octonion algebra and Clifford algebra $Cl(8)$ to Standard Model structure. Gillard-Gresnigt (arXiv:1906.05102) demonstrate that minimal left ideals of $Cl(8)$ contain three generations of Standard Model fermion states with full Lorentzian, chiral, weak isospin, spin, and electrocolor degrees of freedom, with the triality automorphism of $Cl(8)$ underlying the extension from one to three generations.

The framework's structural claims are consistent with these research programs and may provide structural motivation for their algebraic constructions. The framework does not by itself complete the octonion approach. The community consensus is that complete anomaly-free Standard Model derivation has not yet been achieved by any single program. Specific remaining issues include right-handed neutrinos, chiral coupling generation, and unique determination of hypercharge values.

The self-nesting closure

Item 2 admits a structural closure that is internal to the framework. Item 2 asks why specific gauge structure exists. The framework's answer is recursive permutation of generative ground producing local instances that mediate back to universal completion. If Item 2 is itself an instance of that recursive permutation, then gauge structure is what Shape Zero looks like at the scale of particle physics.

The framework derives gauge structure by being gauge structure at the appropriate level of nesting. The Standard Model is what the framework appears as when it asks about itself at the scale of fundamental particles. Self-similarity is strict: each level contains the same pattern that produced it. Particle physics, atomic structure, biological organization, cosmological structure are nesting levels of the same recursive structure.

This closure is bounded by the calculator's capacity, not by the framework's content. As far as any specific instance can compute the framework, it works. Beyond that boundary, the framework opens because completion would require an outside vantage that no instance possesses. The framework's claim is that this is what reality actually does — produces specific structure at every scale through the same mechanism, with the mechanism being identifiable from any of its instances.

Honest scope of this closure

Mathematicians who work on self-referential systems would recognize this as a legitimate closure. Physicists who require derivations from outside the system would consider it a dodge. Both reactions are defensible.

What the closure does: it makes the framework genuinely self-contained. What's needed for further progress is more nesting (deeper computation by more calculators) rather than additional external machinery.

What the closure does not do: it does not produce specific predictions for masses, coupling constants, or generation mixing. Those require the connection to active

research programs (octonion approach, lattice gauge theory simulation, etc.) to advance through their own work.

Minimal computational test protocol (new in v5.1)

The cheapest decisive step for Item 2 is a numerical experiment executable on a single GPU or CPU. The U(1) claim predicts a Peierls-type synthetic gauge phase generated by the velocity coupling $\beta \cdot \dot{x}_j$. Protocol:

Step 1 — Ring construction. $N = 64$ phi-attractor oscillators on a ring, nearest-neighbor coupling $c_{ij} = c$, all oscillators initialized at rest on the φ attractor, drive $\varepsilon = 0$ (the drive is not needed for the linearized test and removing it isolates the gauge signature).

Step 2 — Linearization. Perturb one site by a small δx and evolve. In the linearized regime about φ , the network is a lattice of harmonic modes with effective stiffness $\sqrt{5}$ and a nonreciprocal hopping term proportional to β .

Step 3 — Holonomy measurement. Launch counter-propagating wave packets (clockwise and counterclockwise) and measure the accumulated phase of each after one full loop, via the analytic signal (Hilbert transform) at a fixed probe site.

Falsifiable prediction: the clockwise-counterclockwise phase difference $\Delta\theta$ is nonzero for $\beta \neq 0$, scales linearly in β for small β , and vanishes identically at $\beta = 0$.

Equivalently, the linearized band structure shifts $k \rightarrow k - \beta A$ with A the loop-averaged synthetic potential. If $\Delta\theta = 0$ for all β , the U(1) emergence claim fails as stated.

Analytical caveat on the coupling form (v5.1). Linearizing about φ with symmetric nearest-neighbor coupling and the $\beta \cdot \dot{x}_j$ term as written gives the plane-wave dispersion $-\omega^2 - i(\gamma + 2c\beta \cos k)\omega + \sqrt{5} + 2c(1 - \cos k) = 0$: the β term enters as k -dependent damping, even under $k \rightarrow -k$, which produces no Peierls phase. A genuine synthetic vector potential requires directionality. The minimal modification is the antisymmetric (gyroscopic) form $\beta(\dot{x}_{\{i+1\}} - \dot{x}_{\{i-1\}})$, which contributes $2c\beta\omega \sin k$ — odd in k — and shifts the small- k dispersion as $k \rightarrow k - \beta\omega$, i.e. a bona fide synthetic potential $A \propto \omega$. The protocol should therefore run both forms: the symmetric form is predicted to give $\Delta\theta = 0$ (control), the antisymmetric form $\Delta\theta \propto \beta$ (test). If the framework's structural claims require the symmetric form specifically, the U(1) emergence claim needs restatement.

Step 4 — Non-Abelian extension. Repeat on a dimerized ring (SU(2) claim): transport a two-component excitation around two distinct loops in each order and compare final states. Prediction: the two orderings disagree (non-commuting Wilson loops) when the block coupling is non-diagonal, and agree when it is diagonal.

Numerical results (v5.2)

The protocol above was executed in July 2026 on the full nonlinear network (no linearization in the dynamics; the linear regime was selected by amplitude where stated). Every quantitative prediction below was stated before it was measured.

Fourteen self-contained scripts accompany this document: `phi_gauge_test.py`, `phi_gauge_nonlinear.py`, `phi_gauge_closure.py`, `phi_gauge_wilson.py`, `phi_gauge_chiral.py`, `phi_gauge_delta.py`, `fano_reduction.py`, `phi_gauge_precession.py`,

phi_gauge_decaymap.py, hk_alignment_convexity.py, s1_gradient_flow.py, s2_universality.py, s2b_winding.py, s3_roles.py (numpy only; each runs in about one minute on commodity hardware).

Result 1 — U(1) confirmed; symmetric form falsified. $N = 64$ ring, dispersion asymmetry $\Delta\omega(k) = \omega(k) - \omega(-k)$ measured by space-time FFT. Antisymmetric (gyroscopic) coupling: $\Delta\omega$ at $k = \pi/2$ measured $-0.038, -0.098, -0.200$ against theory $-0.040, -0.100, -0.200$ for $\beta = 0.02, 0.05, 0.10$; the $\sin k$ dependence confirmed at $k = \pi/4$ (-0.139 vs -0.141). Symmetric coupling (the form originally written in the spec): $\Delta\omega = 0.0000$ at every β , and with light damping the network is parametrically unstable for $2c\beta > \gamma$ — k -dependent anti-damping, not a gauge field. The caveat of the v5.1 coupling-form analysis is thereby confirmed on both branches: the U(1) claim holds for the antisymmetric form and fails for the symmetric form.

Result 2 — nonlinear protection with derived correction law (corrected in v5.3). Amplitude sweep at $k = \pi/2$ with single traveling waves and phase-slope frequency estimation. Second-order (Lindstedt) perturbation theory gives the closed-form nonlinear dispersion $L(k, \omega) = (A^2/4)(4/\sqrt{5} + 2/L(2k, 2\omega))$ with $L(k, \omega) = -\omega^2 + \sqrt{5} + 2c(1 - \cos k) - 2c\beta\omega \sin k$, which reduces exactly to the textbook quadratic-oscillator shift $-5a^2/12\omega_0^3$ at $c = 0$. It predicts a band-centre softening of $-0.0973 \cdot A^2$ and a gauge-asymmetry correction of $+0.0349 \cdot \beta \cdot A^2$, both confirmed by measurement with corrected wave preparation (exact β -dependent frequency in the initial velocity field; two-term A^2/A^4 fits): softening -0.0973 , correction coefficient 0.0349 at $\beta = 0.025$ and 0.050 , three-digit agreement. The protection mechanism stands: $\omega^2 + 2c\beta \sin k \cdot \omega - \Omega^2 = 0$ yields $\Delta\omega = -2c\beta \sin k$ for any direction-blind Ω^2 ; the correction arises solely from the β -dependence of the second-harmonic denominator. Combined law at $k = \pi/2$: $\Delta\omega = -2c\beta \sin k \cdot (1 - 0.0175 \cdot A^2)$.

Correction of the v5.2 value. v5.2 reported $\delta = -0.0598 \cdot \beta$. That number is retracted as a preparation bias: the v5.2 initial condition set the velocity field with the $\beta = 0$ frequency, seeding an $O(\beta)$ counter-rotating admixture in the measurement bin whose nonlinear frequency pulling is itself odd and linear in β — so the artifact passed the parity check (odd to 10^{-14}) and the linearity check (1.4%) that were designed to catch spurious effects. The bias is reproduced and isolated in phi_gauge_delta.py: identical integrator and estimator, only the initial velocity frequency differs ($-0.060 \cdot \beta$ with the v5.2 preparation, $+0.0349 \cdot \beta$ with the corrected one). Recorded as a caution: parity and linearity checks do not exclude artifacts that share the symmetry of the effect under test.

Result 3 — direction-selective parametric decay (discovered). An apparent anomaly at $\beta = 0.075$ resolved into new physics: a resonant four-wave decay $k_0 + k_0 \rightarrow 0 + \pi$ of the traveling carrier. Verified three ways: the decay products appear in exactly the predicted bins oscillating at their (softened) linear frequencies ($1.466 \approx \Omega(0)$, $2.471 \approx \Omega(\pi)$); frequency matching is exact at finite amplitude (product sum 3.9375 vs softened carrier $2\omega_+ \approx 3.937$); and linear theory locates the resonance crossing at $\beta^* = 0.0629$, where it was found. The decay is nonreciprocal: at identical β and amplitude, the $+k$ carrier's decay product grows exponentially (mode retention falling to 0.797) while the $-k$ carrier is fully protected (retention 1.000) — the synthetic gauge field tunes one propagation direction into the resonance and the

other out of it. The gauge field does not merely shift the band; it selects which direction is permitted to decay.

Result 4 — non-commutative Wilson-loop transport confirmed. Dimerized ring (2-component internal space per node), orthogonal link matrices, $N = 128$, wave-packet transport in the linear regime where internal $O(2)$ symmetry is exact. Two links — a 45° rotation and a reflection — traversed in both orders (same link multiset, only path-ordering differs): final polarizations 135.00° vs 44.95° , a 90.06° splitting against a predicted 90° , with per-configuration agreement to 0.05° . Abelian control (two rotations): orderings identical to 0.01° . Polarization purity ≥ 0.998 in all runs, confirming that each link is locally pure gauge (no scattering) and the path-ordered holonomy is the only physical invariant. The holonomy group realized is $O(2)$ — non-Abelian via the rotation/reflection mix — rather than $SU(2)$ proper, which requires the complexified (4-real-component) dimer; the property Step 4 tests, non-commutativity of transport, is established.

Scope of what is now established. The load-bearing sub-claim of Item 2 — that the phi-attractor network genuinely carries synthetic gauge fields of the claimed types — is confirmed for $U(1)$ (generated dynamically by the gyroscopic coupling) and for non-Abelian transport (with link matrices imposed lattice-gauge style). The derivation gap is now precisely localized: show that dimerized velocity coupling dynamically generates non-commuting link transport, rather than having it written in. A further open observation: the on-site u^2 term breaks internal $O(2)$ symmetry at second order (it acts per component, not on $|u|^2$), so the non-Abelian gauge symmetry is exact only in the linear regime; at finite amplitude the nonlinearity selects a preferred internal frame, predicting slow polarization precession — untested.

A positive result does not establish Standard Model gauge structure; it establishes that the network genuinely carries synthetic gauge fields of the claimed type, which is the load-bearing sub-claim.

Numerical results — dynamical generation of non-Abelian transport (v5.3)

Result 5 — non-Abelian transport dynamically generated; no link matrices imposed. The complexified dimer experiment (script `phi_gauge_chiral.py`) closes the gap identified in v5.2. Each node carries $u \in \mathbb{R}^4$ (two phi-dimers). An intra-node gyroscopic term $\kappa \mathbb{J} \dot{u}$ dynamically selects the complex structure — the node spectrum splits into chiral modes, and a chirality-pure wave packet is a $\mathbb{C}^2$ spinor field with no complex structure imposed by hand. Inter-node transport arises solely from the directional velocity coupling $c \cdot W(v_{n+1} - v_{n-1})$ with constant real matrices W in the equations of motion — the matrix generalization of the scalar gyroscopic coupling of Result 1. No link, transport, or holonomy matrices appear anywhere in the model.

Structural result: the conservative class is the gauge class. For matrix-valued directional velocity coupling the zero-power (energy-conserving) condition reverses the scalar rule: the telescoping argument requires W symmetric, not antisymmetric. Real symmetric matrices commuting with the emergent complex structure $\mathbb{J}$ correspond exactly to Hermitian complex 2×2 matrices — precisely the $u(2)$ gauge-potential class required for unitary transport. Energy conservation and gauge

structure independently select the same coupling family. The scalar U(1) case of Result 1 is recovered as the trace (σ_0) part.

Measurements. $N = 200$ ring, chirality-pure packet at $k = \pi/2$ ($\omega = 1.8233$), two coupling regions with $W \propto \sigma_x$ ($\approx 90^\circ$) and σ_y ($\approx 60^\circ$), traversed in both orders. All transport predictions were computed by an independent Bloch-theory module (per-branch wavenumbers at the conserved packet frequency, per-link WKB accumulation). Spinor Bloch vectors agree with prediction to $\leq 0.05^\circ$ at every measurement stage; path-ordering splitting between traversal orders: 59.86° measured vs 59.84° predicted; Abelian control (parallel axes): 0.00° . Chirality purity 0.984 throughout; energy drift 5×10^{-7} over the full run, confirming the symmetric-W conservativity claim in the dynamics itself. Mechanism-level validations: per-component wavenumbers measured inside a coupling region ($1.6569 / 1.4904$) match the two-branch dispersion ($1.6579 / 1.4837$), and the downstream site-wise inter-component phase ($+0.6807$ rad) matches the boundary-matching prediction $+\Delta k \cdot L$ ($+0.6860$ rad).

Record of errors found during this experiment. Two errors were found and localized during the run, both in the theory and readout modules and neither in the simulated physics: a chirality-branch sign error in the prediction dispersion (exposed by a uniform measured/predicted angle ratio equal to the ratio of the two branch frequencies) and a transposed density matrix in the spin readout (exposed as a global n_y sign flip and pinned by exact eigenproblem, in-medium wavenumber, and downstream-phase measurements). They are recorded because the diagnostic chain — three independent measurements progressively excluding every physical explanation — is itself evidence of the result’s robustness.

Scope after v5.3. The v5.2 derivation gap (“link matrices imposed by hand”) is closed: non-commuting transport is generated by the dynamics from constant conservative couplings, and the coupling class is derived (conservativity + compatibility with the dynamically selected complex structure $\Rightarrow u(2)$). The remaining phenomenological freedom, stated honestly: which Hermitian generator acts on which links — the spatial axis profile — is chosen, not derived. The gap has moved from the existence of the structure to the selection of a particular configuration within it.

Secondary measurements (v5.3)

Result 6 — finite-amplitude spinor self-precession, predicted then measured.

The per-component u^2 nonlinearity is not U(2)-invariant: it acts as independent self-phase modulation on each dimer. Harmonic balance for a circular dimer wave (channels DC, co-rotating ($2k, 2\omega$), counter-rotating ($-2k, -2\omega$), the last two split by κ) gives $\Delta\omega = a^2[-1/\sqrt{5} - 1/(4L_+) - 1/(4L_-)]/(2\omega + \kappa)$, predicting spinor Bloch precession about the dimer-population axis at rate $\Omega = C \cdot A^2 \cdot n_z$ with $C = -0.0896$ at $k = \pi/2$ — stated before measurement. Measured (phi_gauge_precession.py): $C = -0.0899$ (0.3%); n_z conserved to three decimals in every run; exact odd symmetry across the equator; A^2 scaling to $A = 0.2$. Interpretation: at finite amplitude the transport picture is the $u(2)$ connection plus a universal derived on-site precession — an amplitude-induced Zeeman-like term about σ_z . At the amplitudes of Result 5 ($A \sim 10^{-3}$) the effect is $\sim 10^{-8}$ and negligible, confirming the internal consistency of the transport measurements.

Result 7 — the nonreciprocal decay window, mapped and located by theory. A 56-point retention map over $\beta \in [0.03, 0.10] \times A \in [0.10, 0.40]$ (`phi_gauge_decaymap.py`; $T = 400$, seeded 10^{-6}). The $+k$ carrier decays only inside a window centred at $\beta \approx 0.070$ – 0.074 for $A \geq 0.30$ (retention down to 0.61), while the $-k$ carrier retains 1.000 at every spot check — directional protection across the whole plane. The v5.2 puzzle (decay observed at $\beta \approx 0.075$ versus a naive resonance at 0.0629) is resolved quantitatively by the derived $O(A^2)$ theory: the pump softening ($-0.195A^2$ on $2\omega_{\text{pump}}$) is overcompensated by the cross-softening of the product modes in the pump field (DC stiffness shift plus mixing sidebands, $-0.322A^2$ on $\omega(0) + \omega(\pi)$), giving $\beta_{\text{res}}(A) = 0.062 + 0.064A^2 - 0.069$ at $A = 0.30$ and 0.074 at $A = 0.40$, matching the map within grid resolution. A hand-arithmetic factor error briefly suggested a flat window during analysis; the exact evaluation, which supersedes it, is what matches the data.

Universality control and the status of the golden ratio (v5.3)

Universality (S2). All measured dynamical laws were rerun on a generic-cubic control lattice (stiffness $K = 2$, nonlinearity $\alpha = 0.7$, replacing $\sqrt{5}$ and 1). Every law reproduces under the generalized closed form $L(k, \omega) = (A^2/4)\alpha^2(4/K + 2/L_2)$: the linear gauge law $\Delta\omega = -2c\beta \sin k$ is literally stiffness-free (measured -0.10000 on both lattices); self-softening -0.0551 measured versus -0.0551 predicted; asymmetry correction $+0.0197$ versus $+0.0196$, with the ϕ -lattice's $+0.0349$ and the control's $+0.0196$ revealed as two points on the single law $4\alpha^2/L_2^2$; the decay window relocated by the same cross-softening theory (deepest retention 0.528 at $\beta = 0.074$ versus predicted centre 0.0754, $-k$ protected). Conclusion, stated deliberately and in advance of any referee: the numerical value $\sqrt{5}$ is decorative in the dynamics layer. Nothing measured so far requires the golden ratio as a constant.

Where the golden ratio actually lives. The framework's ϕ was never a chosen constant: it is the output of a parameter-free thought experiment. Begin from zero; observing zero makes it one thing; the observation implies observer and observed, two; thereafter each stage is the present whole related back to the prior whole — the Fibonacci recurrence. The superposition clause (still zero, yet one) is the fixed-point equation $x = 1 + 1/x$: the whole as its unit plus its reciprocal relation to origin. Its continued fraction is all ones — the unique expansion that introduces no multiplicities — so ϕ is the fixed point of self-reference with nothing added. The same all-ones expansion is, by the theory of Diophantine approximation, the extremal case of resistance to rational approximation (Hurwitz: $|\phi - p/q| \geq c/(\sqrt{5} q^2)$ with $\sqrt{5}$ optimal — the same $\sqrt{5}$), and in KAM theory the golden torus is the last destroyed. Minimal generative structure and maximal endurance are one mathematical object read in two directions. The framework's claim is therefore relocated: not “ ϕ in the potential” but “endurance selects maximally irrational winding, whose extremum is ϕ .”

First measurement of the selection mechanism (S2b). Two carriers were run jointly on the ϕ -lattice and the survival landscape swept against the winding ratio $\rho = \omega_2/\omega_1$ across 28 ring wavenumbers. The prediction hierarchy was stated before running — three-wave dispersion geometry expected to dominate, Diophantine structure a doubtful residual — and the data inverted it: survival correlates at Spearman $+0.66$ with the Diophantine badness of ρ and at -0.13 with three-wave

mismatch (all three-wave channels sit uniformly off-resonance at this carrier and are inert). The catastrophic dips (survival 0.05–0.06) bracket the 1:1 tongue; a carrier pair sitting near 7/6 survives fully, consistent with tongue widths collapsing at higher order; the high-irrationality side is a quasiperiodic plateau. Cautions recorded honestly: single carrier, single amplitude, single horizon; the correlation is partly carried by the dominant 1:1 tongue; this establishes that rationality-distance organizes endurance in this lattice — the prerequisite for noble selection — and does not yet establish φ -specific fine structure, which is the designed follow-up.

Specific remaining technical work

Derive coupling matrices c_{ij} from framework's structural claims rather than choosing them to produce desired gauge groups.

Show that framework structure satisfies Dynamic Balance Möbius symmetries rigorously rather than analogically.

Reduced in v5.3: forcing the Fano plane now requires exactly one further axiom beyond triadic closure — equivalently “any two triads meet” or “every element lies in exactly three triads” (see the reduction theorem above; AG(2,3) is the counterexample without it). Remaining: motivate one of the two equivalent forms from Rank 2/Rank 3 structure, or accept Fano selection as conditional.

Executed in v5.2 (Section 5.2.8) and v5.3 (Result 5): the complexified dimer, the dynamical generation of non-commuting transport, and the analytic derivation of the amplitude correction (Result 2, corrected) are done. The finite-amplitude precession and the decay-window map are done (Results 6 and 7). Remaining item from Result 5: derive the spatial axis profile of the $u(2)$ coupling (which generator on which links) from structural claims rather than choosing it.

Engagement with the octonion approach as collaborative research direction rather than imported solution.

Specialist profiles

For the dynamics and gauge emergence: nonlinear dynamics specialist with experience in coupled oscillator networks, ideally familiar with synthetic gauge field emergence in classical systems.

For the octonion connection: mathematical physicist working in division algebras, Clifford algebras, or the Furey/Dixon program.

For the self-nesting closure: mathematician or philosopher working on self-referential systems, category theory of self-similar structures, or higher topos theory.

5.3 Item 3: Sufficient Condition for Lyapunov Stability

Status: Closed at framework level

Resolution

The Sobolev space H^s on a compact Riemannian manifold has spectral characterization via Laplace-Beltrami operator:

$$\widehat{H^s}(M) = \{f = \sum f_k M_k : \sum (1+\lambda_k)^s |f_k|^2 < \infty\}$$

with reproducing kernel $K_s(x,y) = \sum (1+\lambda_k)^{-s} M_k(x) M_k(y)$. LB eigenbasis EDMD with this kernel is kernel EDMD with Sobolev RKHS structure. Bold-Schaller-Schimperna-Worthmann proportional error bounds (arXiv:2501.08709) and Philipp-Schaller-Worthmann-Peitz-Nüske kernel EDMD framework (arXiv:2312.10460) apply.

Result: residual $R(\sigma)$ is second-order near σ^* with proportional bound $L_R \cdot |\sigma|$. Koopman contraction grows linearly with $\gamma \cdot \lambda_{\text{gap}} \cdot |\sigma|$. The sufficient condition holds for σ within radius $r_c = \gamma \cdot \lambda_{\text{gap}} / L_R$.

For intersubjective coupling, the Lyapunov-based small-gain theorem for infinite networks via Mironchenko-Kawan-Glück (arXiv:2007.05705) provides ISS framework. Schur complement and Petersen's lemma deliver the LMI argument.

Specific remaining technical work

Verify proportional error bound theorem applies to Sobolev kernel from LB eigenbasis on Fisher manifold. Compute explicit residual constant L_R and spectral gap λ_{gap} . Construct augmented LMI yielding critical η threshold.

6. Bidirectional Reading and Self-Reproduction

6.1 The Two Directions

The framework can be read in two directions. Read upward, it describes structure forming through recursive permutation — how complexity emerges from simpler foundations through phi-governed recursion. This reading produces hope: the building can continue.

Read downward, it describes structure dissolving through approach to the void — how the universe sinks toward zero with the structures we observe being patterns of the sinking rather than achievements of building. This reading produces dread: the bottom can't be reached but the falling is real.

Both readings are accurate. Building gives infinite bounded complexity — structure that can never complete because each new level generates more relationships than can be tracked at the current level. Falling gives infinite unbounded nothingness — the void that can't be reached because reaching it would mean being it. Neither direction completes. Both are simultaneously true.

The two directions as geometry (v5.1). With the corrected metric of Section 2.4, the bidirectional reading is no longer only interpretive. The Hellinger-Kantorovich geometry canonically decomposes every infinitesimal motion of μ into a transport component — rearrangement of shape at fixed mass, the Wasserstein channel — and a reaction component — creation and annihilation of mass, the Fisher-Rao channel pointing along the ray toward or away from the void. Building is motion in the transport channel; falling is motion in the reaction channel toward μ_0 . The framework's two readings are the two orthogonal channels of its own configuration-space metric. The void-resistance potential acts only on the reaction channel, which is why structure can transform freely while annihilation is resisted.

The configuration's reality is neither building nor falling separately. It is the simultaneous condition of structure forming as it dissolves, with neither the formation nor the dissolution being prior. The 0-1-2-3 hierarchy is not a sequence pointing one direction. It is the configuration that exists as the simultaneous condition of structure forming and dissolving, with the 1 being where formation and dissolution meet, the 2 being how that meeting differentiates, the 3 being how the differentiation maintains itself temporarily, and the 0 being what the whole configuration exists against without ever resolving into.

6.2 Self-Reproduction

Local exhaustion produces stable form, but the universe keeps going. New stars form as old ones die. New cells form as old ones break down. The framework is self-reproducing at the structural level: each completed local instance generates conditions for new local instances because the structure that produced it is the same structure that produces them.

The pattern doesn't end at any local exhaustion because the pattern itself is what makes new local instances possible. This is not infinity in the unbounded sense, but unending in the continuously-regenerating sense. The framework doesn't close because closing would mean ending, and the structure's nature is to not end.

6.3 The Calculator Bound

The framework opens and closes simultaneously because its limits are the calculator's limits, not the framework's limits. As far as any specific instance can compute the framework, it works. The point at which it stops working is the point at which the computation runs out of capacity, not the point at which the framework runs out of content. The framework is always closed at the boundary of what's been computed and always open beyond that boundary.

Most successful theories are closed-and-open in this same way — complete in their applicable domain, incomplete at boundaries where calculation breaks down. General relativity is closed within tested regimes, opens at black hole singularities and the early universe. Quantum mechanics is closed within tested regimes, opens at the measurement problem and quantum gravity. The framework's distinction is articulating this structure explicitly rather than treating boundary openness as embarrassing failure.

6.4 The Search as Instance

If the framework is correct, the search for an answer is itself an instance of what the framework describes. Looking up finds an answer because looking up is one of the ways the framework manifests. Looking down finds an answer because looking down is another. The search produces what it's looking for because the search is the same recursive permutation that produces everything else.

This is not solipsism. It is what self-similar structure means when the seeker is part of the structure being sought. The framework couldn't close from outside because the work of articulating it is already the framework operating. Every refinement is the structure refining its own articulation through one of its local instances.

7. Items Genuinely Open Beyond Verification

$\alpha = d_I/2$ explicit derivation from information dimension. Hypothesis stated, derivation incomplete.

T_{12} flow equation for intersubjectivity dynamics. Structure identified, equation not specified.

Classical/quantum interface term. Conceptually motivated, derivation incomplete. The Slotine-Lohmiller MIT bridge work (Proceedings of the Royal Society A April 2026, DOI:10.1098/rspa.2025.0413) may be relevant.

Specific Standard Model matter content beyond gauge structure (masses, mixing matrices, generation count derivation). Connects to active research in the octonion approach.

The HK length-scale constant c of Section 2.4 is currently a normalization choice. Whether the framework fixes it intrinsically — for example via the ϕ -recursion capacity bound of Section 4 — is open.

8. Established Mathematical Connections

Information geometry (Amari 1985).

Koopman operator theory (Koopman 1931, EDMD via Williams-Kevrekidis-Rowley 2015).

Optimal transport and Wasserstein geometry (Villani 2003, 2008).

Unbalanced optimal transport and the Hellinger-Kantorovich distance (Liero-Mielke-Savaré 2018; Chizat-Peyré-Schmitzer-Vialard 2018).

Ambrosio-Gigli-Savaré gradient flows on metric measure spaces (2005).

HWI inequality (Otto-Villani 2000, rigorous via Gentil et al 2020).

Bakry-Émery curvature-dimension condition and RCD spaces.

Kernel EDMD proportional error bounds (Bold et al 2025, Philipp et al 2023).

Lyapunov small-gain theorems for infinite networks (Mironchenko-Kawan-Glück, Kawan et al).

Dynamic Balance principle for golden ratio in NESS (Ruiz 2025).

Synthetic gauge fields in oscillator networks (cold atom physics, photonic lattices, topological metamaterials).

Octonion approach to Standard Model (Furey, Dixon, Gillard-Gresnigt — active research program).

9. Items Corrected During Development

AM/HM mediation operator: empirically tested, found redundant, removed from load-bearing status.

Big Bang parallel for C_Π : discarded. C_Π correctly identified as instance-defined via POVM of local observer.

Banach contraction proof of joint fixed point: incorrect because Koopman is an isometry. Replaced with Lyapunov stability.

Lipschitz Φ_U via direct EDMD transfer: incorrect. Replaced with kernel EDMD proportional error bound via Sobolev RKHS equivalence.

Earlier Item 2 closure attempts via $\text{Spin}^c(3,1)$ constraints, modular symmetry promotion, complexity minimization arguments: identified as overclaim. Item 2 reframed with structural account plus self-nesting closure.

Earlier reference to fringe Photrino paper (Fouad, Applied Physics Research): removed.

Dark matter, dark energy, pitch angle predictions: removed from load-bearing structure.

v5.3: consolidated verification program record added as Section 9b, mirroring the action log.

v5.1: W_2 distance to the zero measure identified as undefined (no admissible coupling exists between measures of unequal mass). Void-resistance potential and ϕ coordinate restated on the Hellinger-Kantorovich geometry over $M(\Omega)$; functional form $V = g \cdot \phi^{-2}$ preserved, Item 1 analysis carried over, exclusion radius reinterpreted as a mass floor.

v5.1: Fano plane selection restated as conditional on the projective incidence axioms, which Rank 2/Rank 3 structure has not yet been shown to entail; previous wording (“forces”) was an overclaim in that respect.

v5.2: numerical program of Section 5.2.7 executed in full; results folded in as Section 5.2.8. The symmetric velocity-coupling form is falsified as a gauge field (zero dispersion asymmetry; parametric instability for $2c\beta > \gamma$) and the $U(1)$ claim is carried by the antisymmetric form, as anticipated by the v5.1 caveat.

v5.2: the “report $\Delta\theta(\beta)$ ” work item is closed; the derivation gap for non-Abelian structure is relocalized from “show gauge structure exists” to “show the link matrices are dynamically generated rather than imposed”.

v5.3: chiral-dimer experiment executed (Result 5). Non-Abelian transport is dynamically generated — no link matrices imposed; path-ordering splitting 59.86° vs 59.84° predicted; Abelian control 0.00° . Structural finding: energy conservation requires symmetric coupling matrices for the directional velocity form, and symmetric + complex-structure-compatible = Hermitian = $u(2)$; the conservative class and the gauge class coincide.

v5.3: the imposed-by-hand gap of v5.2 is closed. Remaining phenomenological freedom restated: the coupling class is derived, the spatial axis profile is chosen.

v5.3: role-theoretic derivation of the Fano supplement. Two postulates lifted from the genesis derivation (role completeness; role minimality) entail supplement (B) and

hence the Fano plane uniquely; the role structure is constructed explicitly (König) and AG(2,3) is excluded by pigeonhole. The Rank $\Rightarrow$ Fano entailment is reduced from an unstated assumption to two named, examinable postulates.

v5.3: universality control executed (S2): all dynamical laws reproduce on a generic-cubic lattice under the generalized closed forms; $\sqrt{5}$ declared decorative in the dynamics layer. The ϕ -claim relocated to its two defensible legs: parameter-free origin (all-ones continued fraction) and endurance selection (Hurwitz/KAM identity). First measurement of the selection mechanism (S2b): survival organized by Diophantine badness of the winding ratio (Spearman +0.66) with three-wave geometry inert (−0.13) — stated prediction hierarchy inverted by the data.

v5.3: Item 1 dynamically validated. The corrected local claim was run as an actual HK gradient flow: convergence at the derived rate to four digits (the bound is sharp), permanent metastability beyond the horizon at the marginal mass, basin boundary = horizon to eight digits.

v5.3: Item 1 global convexity claim withdrawn. Counterexample: HK² between Diracs is a minimum of mechanism costs and acquires a concave kink where the base distance crosses the $\pi/2$ transport-reaction horizon, so the alignment term is not λ -convex for any λ along plain HK geodesics, already for $\Phi_U = \text{id}$ (verified numerically to five digits). Claim restated as local to the transport-connected phase, where Dirac-sector convexity is exact; open lemma sharpened to generalized-geodesic convexity within the horizon, with a confinement condition named for Φ_U .

v5.3: secondary measurements executed. Result 6: finite-amplitude spinor self-precession $\Omega = C \cdot A^2 \cdot n_z$ predicted in closed form ($C = -0.0896$) and measured ($C = -0.0899$, 0.3%). Result 7: the nonreciprocal decay window mapped over 56 points; window location $\beta_{\text{res}} = 0.062 + 0.064A^2$ derived from pump self-softening plus product cross-softening, resolving the v5.2 offset puzzle; $-k$ protection verified across the plane.

v5.3: Fano reduction theorem. Triadic closure alone is shown NOT to entail the Fano plane (counterexample AG(2,3), machine-verified); forcing PG(2,2) requires exactly one further axiom, in equivalent global (any two triads meet) or local (every element in exactly three triads) form. The v5.1 conditional wording is retained with the condition now minimal and explicit.

v5.3: Item 1 second-variation appendix supplied. Key simplification: total mass — hence ϕ^2 — is exactly quadratic along HK geodesics (cone-space computation), so the second variation of the void-resistance potential is an identity and the $|\phi''|$ work item flagged since v5.1 is dissolved. R_{void} confirmed exact and conservative; the mass floor is now derived. The open lemma of Item 1 is relocated to the geodesic convexity of the alignment term.

v5.3: Result 2 corrected. The amplitude correction to the gauge asymmetry is $+0.0349 \cdot \beta \cdot A^2$, now derived in closed form from second-order perturbation theory and confirmed by measurement with corrected wave preparation. The v5.2 value $\delta = -0.0598 \cdot \beta$ is retracted as a preparation bias that was odd-linear in β and therefore passed the v5.2 parity and linearity checks. The band-centre softening is likewise

corrected from -0.101 to the derived -0.0973 (A^4 contamination of the single-term fit).

9b. Verification Program Record (July 2026)

Consolidated record of the executed program, mirroring the working action log. Every quantitative outcome is reproducible from the accompanying scripts; predictions were stated before measurement throughout.

Item	Outcome
Result 1 — U(1) gauge field	$\Delta\omega = -2c\beta \sin k$ confirmed to $\sim 1\%$; symmetric scalar form falsified (zero asymmetry + parametric instability)
Results 2-3 — protection & decay	Gauge asymmetry pinned under nonlinearity; direction-selective decay $k_0+k_0 \rightarrow 0+\pi$ discovered, $-k$ protected
Result 4 — Wilson loops	Path-ordering splitting 90.06° vs 90° predicted; Abelian control 0.01° (links imposed at this stage)
A1 / Result 5 — chiral dimers	Non-Abelian transport dynamically generated, no link matrices; ordering 59.86° vs 59.84° ; conservativity $\Leftrightarrow$ u(2) coincidence
A2 — analytic δ	Closed-form PT derived and confirmed ($+0.0349\beta A^2$, softening $-0.0973A^2$); v5.2 value -0.0598β retracted as preparation bias
A3 — second variation	φ^2 exactly quadratic along HK geodesics ($F'' = 2L^2$); V'' an identity; mass floor derived; φ'' obstacle dissolved
A4 — Fano reduction	Triadic closure alone does not entail Fano (AG(2,3) counterexample); one further axiom, two equivalent forms, necessary and sufficient
A5 — secondary measurements	Spinor self-precession $\Omega = C \cdot A^2 \cdot n_z$: $C = -0.0896$ predicted, -0.0899 measured; decay window mapped, $\beta_{\text{res}} = 0.062 + 0.064A^2$
A6 — alignment convexity	Global claim falsified (concave kink at the $\pi/2$ horizon, jump matched to 5 digits); local Dirac-sector convexity exact; open lemma sharpened
S1 — gradient-flow endurance	PASS: rate 1.8529 vs 1.8527 (bound dynamically sharp); metastable mass $= \sqrt{(g/\delta)}$; basin edge $= \pi/2$ to 8 digits

S2 — universality control	All laws reproduce on a generic-cubic lattice; $\sqrt{5}$ decorative in the dynamics layer; φ -claim relocated to origin + endurance selection
S2b — winding selection	Survival organized by Diophantine badness of ω_2/ω_1 (Spearman +0.66); three-wave geometry inert (−0.13); stated hierarchy inverted by data
S3 — role derivation	Role completeness + role minimality $\Rightarrow$ supplement (B) $\Rightarrow$ Fano uniquely; role structure constructed (König); AG(2,3) excluded by pigeonhole
C1 — package freeze	Fourteen reproduction scripts, one command per result; PDF; specialist cover notes; zero-fit experimental predictions addendum

Post-freeze research (the externalization programme E1, targeting v5.4) is tracked in the working action log.

10. Engagement Note

This document is offered for genuine technical engagement. The verification items are stated specifically enough that disagreement should also be specific. Honest negative results are valued equally with positive ones.

The author is an independent researcher working under significant resource constraints. Collaborative engagement that closes verification items, advances open problems, or definitively establishes that they cannot be closed in the form stated, is welcomed.

Specialists in nonlinear dynamics with experience in synthetic gauge field emergence are particularly invited to evaluate the phi-attractor oscillator network, including the minimal computational test protocol of Section 5.2. Mathematical physicists working on the octonion approach are invited to consider whether the framework’s structural claims provide useful motivation for their algebraic constructions. Specialists in unbalanced optimal transport are invited to evaluate the v5.3 form of Item 1: the second-variation appendix, the horizon counterexample, and above all the sharpened open lemma — whether $\text{HK}^2(\nu, \cdot)$ is 2-convex along generalized geodesics within the interaction-connected regime. Mathematicians and philosophers working on self-referential systems are invited to evaluate the self-nesting closure for Item 2.

References

- Amari, S. (1985). *Differential-Geometrical Methods in Statistics*. Springer.
- Ambrosio, L., Gigli, N., Savaré, G. (2005). *Gradient Flows in Metric Spaces and in the Space of Probability Measures*. Birkhäuser.
- Bold, L., Schaller, M., Schimperna, I., Worthmann, K. (2025). Kernel EDMD for data-driven nonlinear Koopman MPC with stability guarantees. arXiv:2501.08709.

Chizat, L., Peyré, G., Schmitzer, B., Vialard, F.-X. (2018). An interpolating distance between optimal transport and Fisher–Rao metrics. *Foundations of Computational Mathematics* 18:1–44.

Furey, C. (2016). Standard model physics from an algebra? PhD thesis, University of Waterloo. arXiv:1611.09182.

Gallouët, T., Monsaingeon, L. (2017). A JKO splitting scheme for Kantorovich–Fisher–Rao gradient flows. *SIAM Journal on Mathematical Analysis* 49(2):1100–1130.

Gentil, I., Léonard, C., Ripani, L., Tamanini, L. (2020). An entropic interpolation proof of the HWI inequality. *Stochastic Processes and their Applications* 130(2):907–923.

Gillard, A. B., Gresnigt, N. G. (2019). The $C\ell(8)$ algebra of three fermion generations with spin and full internal symmetries. arXiv:1906.05102.

Kawan, C., Mironchenko, A., Swikir, A., Noroozi, N., Zamani, M. A Lyapunov-based small-gain theorem for infinite networks. arXiv:1910.12746.

Koopman, B. O. (1931). Hamiltonian systems and transformation in Hilbert space. *PNAS* 17:315–318.

Liero, M., Mielke, A., Savaré, G. (2018). Optimal Entropy-Transport problems and a new Hellinger–Kantorovich distance between positive measures. *Inventiones Mathematicae* 211:969–1117.

Mironchenko, A., Kawan, C., Glück, J. Nonlinear small-gain theorems for input-to-state stability of infinite interconnections. arXiv:2007.05705.

Otto, F., Villani, C. (2000). Generalization of an inequality by Talagrand and links with the logarithmic Sobolev inequality. *Journal of Functional Analysis* 173:361–400.

Philipp, F. M., Schaller, M., Worthmann, K., Peitz, S., Nüske, F. (2023). Error Analysis of Kernel EDMD for Prediction and Control in the Koopman Framework. arXiv:2312.10460.

Ruiz, A. (2025). Dynamic Balance: A Thermodynamic Principle for the Emergence of the Golden Ratio in Open Non-Equilibrium Steady States. *Entropy* 27(7):745. DOI: 10.3390/e27070745.

Slotine, J.-J., Lohmiller, W. (2026). On computing quantum waves exactly from classical action. *Proceedings of the Royal Society A*. DOI: 10.1098/rspa.2025.0413.

Villani, C. (2003). *Topics in Optimal Transportation*. American Mathematical Society.

Williams, M. O., Kevrekidis, I. G., Rowley, C. W. (2015). A data-driven approximation of the Koopman operator. *Journal of Nonlinear Science* 25:1307–1346.

Shape Zero — Working Specification v5.3 — Prepared for sharing with mathematical specialists